

# VIJAYKUMAR KARTHIKEYAN

Los Angeles, CA

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## Education

### University of California, Los Angeles (UCLA)

Sep. 2024 – June 2028

*Bachelor of Science in Bioengineering*

*Los Angeles, CA*

- Technical Breadth in Computer Science + Minor in Data Science Engineering
- Relevant Coursework in Molecular Bioengineering, Machine Learning and Modelling, Biotransport, Thermodynamics, Data Structures & Algorithms, Biomedical Transducers, Biocompatibility, and Biomedical Ethics

### Paul Laurence Dunbar High School

Aug. 2020 – May 2024

*Math Science Technology Center (selective admissions)*

*Lexington, KY*

- GPA - 4.00 | ACT - 35 | 400-hour Capstone Research

## Experience

### UCLA Bioengineering

Jan. 2025 – Present

*Student Researcher @ The Kamariza Lab*

*Los Angeles, CA*

- Studying the mechanisms & activation of CRISPR/Cas13 for nucleic acid diagnostics, under Dr. Mireille Kamariza
- Engineering platforms to design *in vitro* CRISPR assays for infectious disease detection without noncanonical activation

### Cogenesis, Inc.

Mar. 2025 – Nov. 2025

*Cofounder and Chief Technology Officer*

*Los Angeles, CA*

- Led technological development at a biotech startup which incubated in Founders, Inc. and backed by Startup UCLA
- Launched an AI-powered cloud platform that enables biotech and pharma researchers to analyze biological images

### University of Kentucky Center for Applied Energy Research

Oct. 2022 – June 2024

*Student Researcher @ The Risko Lab*

*Lexington, KY*

- Built ML models to forecast Organic Semiconductor interactions for electronics via Dr. Chad Risko

## Projects

### Paper-Based CRISPR-Cas13a RNA Diagnostic for Tuberculosis | *Diagnostics, Microscopy* May 2026 – June 2026

- Engineered a paper-based CRISPR-Cas13a diagnostic strip for point-of-care *Mycobacterium tuberculosis* RNA detection via fluorescent readout. Optimized a two-stage assay protocol using BSA passivation and trehalose stabilization

### Polaris - Agentic Hospital Paging System | *AI Agents, NextJS, FastAPI*

Apr. 2026

- Won Best Social Impact Award at LA Hacks 2026
- Created AI paging system that alerts clinicians

### Kinetiq - Smart Knee Brace to track Recovery | *Swift, React, Django, AWS Cloud*

July 2025 – Present

- Lead Fullstack developer of React Web App and Swift iOS App. Architected REST Backend Application on AWS EC2
- Published iOS application on App Store and Presented Research at 2nd Annual Arizona Digital Health Symposium

## Technical Skills

**Languages:** Python, JavaScript, C++, Swift, PostgreSQL, German (Novice), Tamil (Native), Telugu (Native)

**Developer Tools:** AWS Cloud (EC2, Lambda, S3, etc.), Docker, Google Cloud, REST APIs, React, Claude API

**Laboratory Experience:** In Vitro Transcription, qPCR, Cloning, Bacteria, CRISPR-Cas Diagnostics, PRISM, Benchling

## Leadership / Extracurricular

### Biomedical Engineering Society

Sep. 2024 – Present

*BioHack Chair | Design Team Project Manager*

*UCLA*

- Organized the 48-hour *BioHack* at UCLA with 100+ participants. Partnered with Cognition AI and Revilico, Inc.
- Managing 10+ members to develop kinesiology tape capable of muscular fatigue monitoring in athletes

### VEST @ UCLA

Jan. 2025 – Present

*Internal Vice President*

*UCLA*

- Managed internal club direction by organizing projects & events. Built AI content generation pipeline for lookbk.com